

Unmasking Plan Performance

A Practical Guide to Using GATHER_PLAN_STATISTICS for Oracle Optimization

The Estimates vs. Reality

Why predictions alone are never enough.

Key Performance Indicators

E-Rows (Estimated)

Derived from data dictionary statistics. It represents the optimizer's best guess of how many rows will be returned at each step.

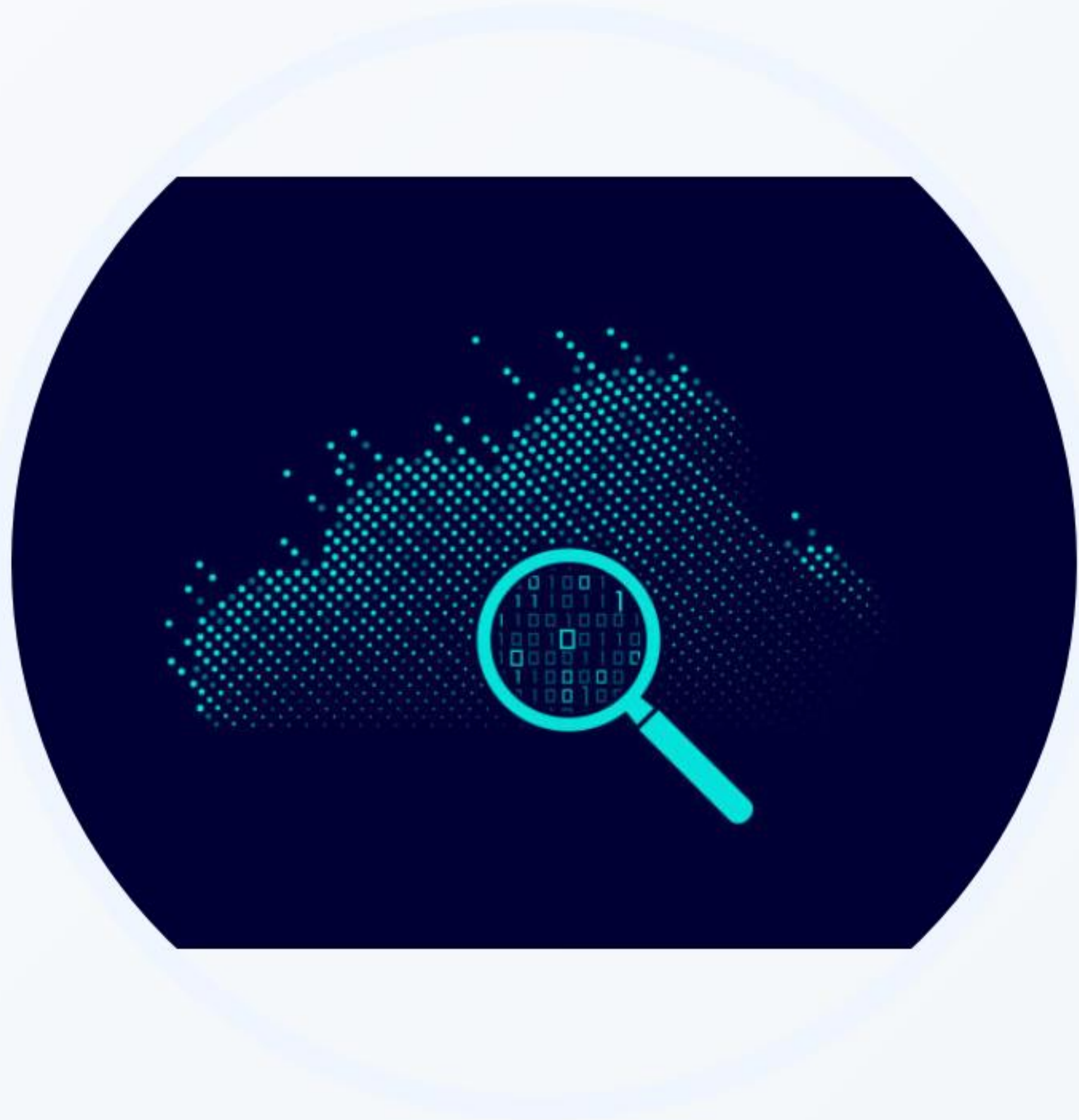
Risk: Can be grossly inaccurate due to data skew or stale statistics.

A-Rows (Actual)

Calculated during actual query execution. It provides the definitive count of records processed by every operation.

Value: Immediately exposes cardinality misestimates and join bottlenecks.

Visualizing the Problem



Identifying the "Hidden" Gap

A query might look perfect in a standard explain plan but fail in production. The magnifying glass must be applied to the **runtime behavior**.

By comparing E-Rows and A-Rows, developers can pinpoint the exact operation where the optimizer's logic diverged from data reality.

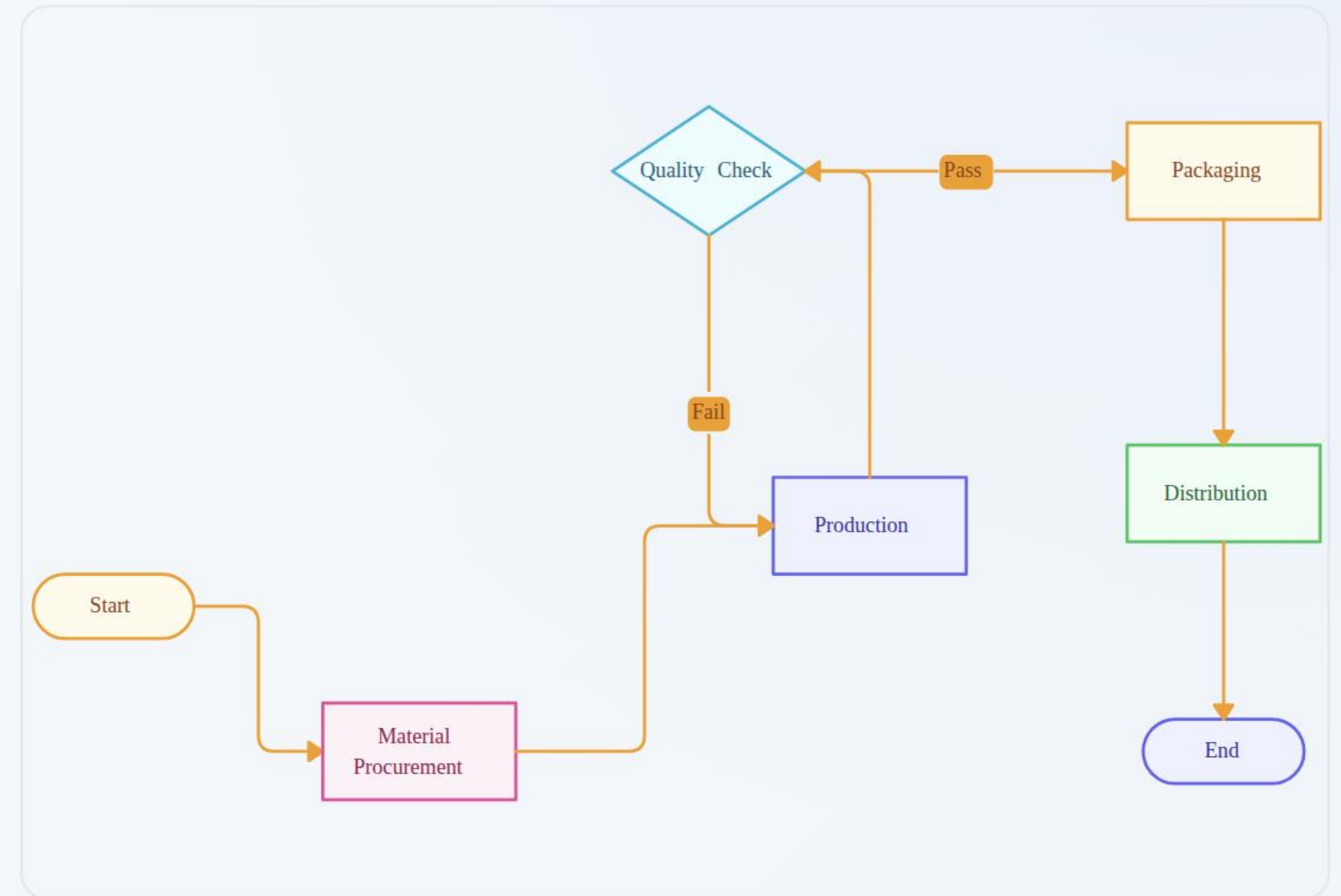
The Solution: Statistics Hint

GATHER_PLAN_STATISTICS

By injecting this hint, you instruct the database engine to instrument the execution pipeline. It collects:

- Actual row counts per operation.
- Number of buffer gets (I/O).
- Actual execution time per step.
- Number of times a loop started.

This is the "Gold Standard" for performance troubleshooting.



Diagnostic Extraction



Inject Hint

Add /*+
GATHER_PLAN_STATISTICS
*/ to the SELECT statement.



Execute SQL

Run the query to completion
in your SQL client or
application.



Pull Plan

Run
DBMS_XPLAN.DISPLAY_CURSOR(format=>'ALLSTATS
LAST').

Comparison Case Study

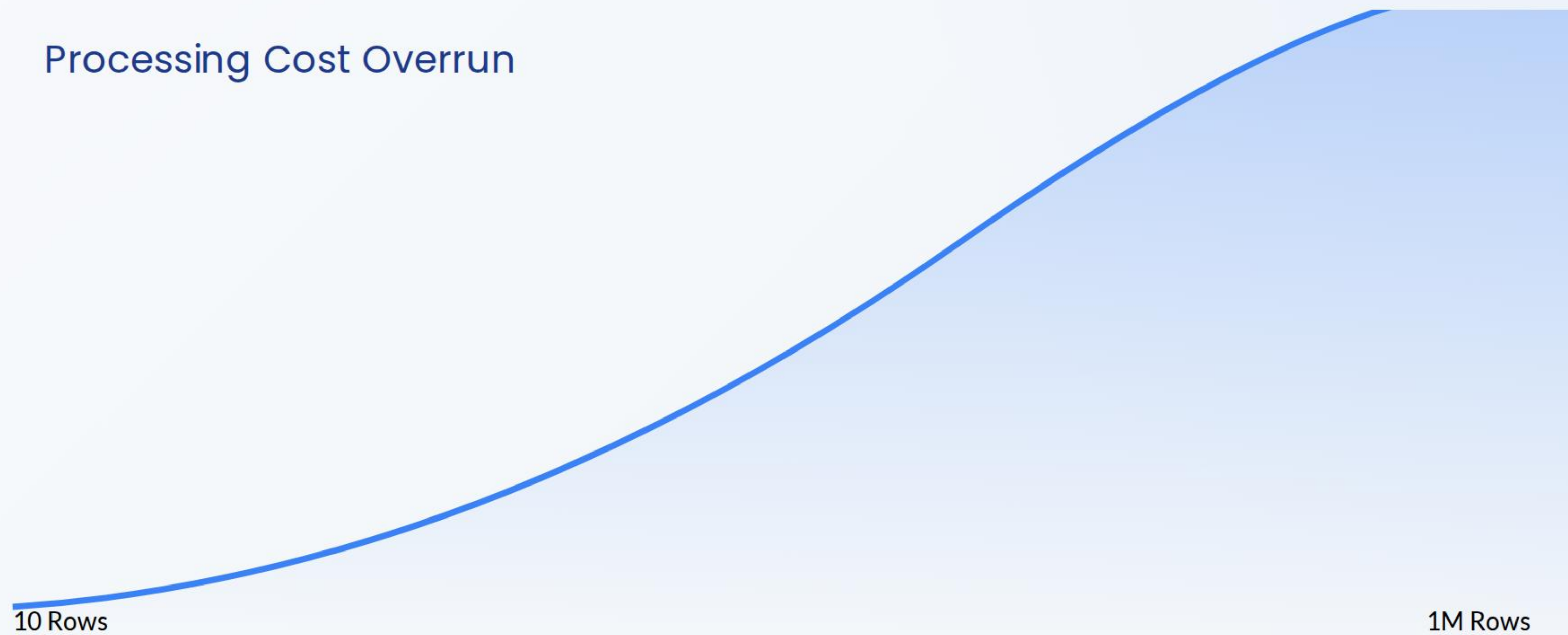
Operation Name	E-Rows (Guess)	A-Rows (Fact)	Diagnosis
TABLE ACCESS FULL (ORDERS)	500,000	1,000	Overestimate (Manageable)
NESTED LOOPS (JOIN)	10	1,000,000	Critical Error (100,000x Gap)
INDEX RANGE SCAN	1	1,000,000	Fatal Latency Hit

Impact of Cardinality Gap



A misestimate of this scale forces Nested Loops where a Hash Join should have been used, causing hours of delay.

Cost Curve Inaccuracy



Permanent Resolution

Fixing the Metadata

Identifying the gap is only the first half. Permanent solutions include:

- **Histograms:** Solve distribution skew.
- **Dynamic Sampling:** Runtime stats.
- **Extended Stats:** Column correlation.
- **SQL Plan Baselines:** Force stability.



The Tuning Workflow



1. Identify

Find long-running queries using AWR, ASH, or Active Session History.



2. Verify

Apply the statistics hint and run the SQL to capture real-time metrics.



3. Analyze

Compare E-Rows vs. A-Rows to find the critical cardinality misestimate.



4. Resolve

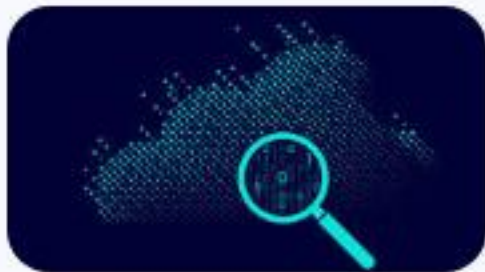
Apply histograms or extended statistics to fix the underlying data model.

Questions?

Start looking at A-Rows today to unlock 10x performance improvements.

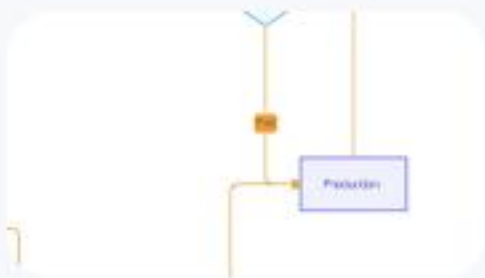
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